

Learning Outcomes

By the end of this lab session, students will be able to

- Construct meaningful new features.
- Select relevant and measure feature importance using different techniques.
- Apply Principal Component Analysis (PCA).
- Identify class imbalance.
- Prepare an optimized dataset for machine learning.

Dataset:

- Load the **Titanic dataset** and apply suitable preprocessing methods.

Part A: Feature Construction

1. Construct two new features from the dataset.
(e.g.: Fare per person can be constructed from: $\frac{Fare}{Family Size}$)
2. Give reasons as to why the new features might be important.

Part B – Feature Selection and Feature Importance

You are preparing the dataset for building a machine learning model. Before training the model, you need to identify which features should be retained and which should be removed.

1. Review the dataset and classify the features into:
 - Relevant features
 - Irrelevant features
 - Redundant features

Remove the irrelevant and redundant features and justify your choices.
2. Perform a chi-square test and/or construct a correlation matrix (heatmap). Identify:
 - Features that are highly correlated with the target variable.
 - Features that are highly correlated with each other.

Based on the correlation analysis, suggest any features that could be removed.

3. Apply Forward Feature Selection using a suitable machine learning model (e.g., Logistic Regression) and display:
 - The selected features.
 - The order in which the features were selected.
 - The model accuracy obtained after feature selection.

Answer the following questions:

- a. How many features were selected?
 - b. Were all features selected? Justify.
 - c. Compare the selected features with the results obtained from the correlation analysis and Random Forest feature importance.
4. Train a Random Forest Classifier and display the feature importance scores.
 - Which feature has the highest importance?
 - Which feature has the lowest importance?
 - Compare the feature importance scores with the correlation analysis. Are the results similar?
5. Based on the results obtained from:
 - **Manual feature selection**
 - **Correlation analysis (Filter Method)**
 - **Random Forest feature importance (Embedded Method)**
 - **Forward Feature Selection (Wrapper Method)**

Select the final set of features that you would use to build the machine learning model. Justify your selection.

Part C – Principal Component Analysis (PCA)

1. Standardize the numerical features. Why is feature scaling important before performing PCA?
2. Apply PCA to the standardized features and display,
 - **Principal components**
 - **Explained variance ratio of each principal component**
 - **Cumulative explained variance**
3. Visualize the explained variance of the principal components.
 - Create a scree plot showing the explained variance ratio of each principal component.
 - Create a cumulative explained variance plot.
 - Identify the point at which approximately 95% of the total variance is retained.
4. Determine the minimum number of principal components required to explain at least 95% of the total variance.

- How many original numerical features were used?
- How many principal components are required?
- How much was the dimensionality reduced?

5. Reduce the dataset using the selected principal components. Display the shape of the dataset before and after PCA.

Part D – Class Imbalance and Data Sampling

The Titanic dataset is used to predict passenger survival. Before training a machine learning model, you need to determine whether the target classes are balanced and apply appropriate sampling techniques.

1. Display the class distribution of the target variable (survived).
 - Is the dataset balanced or imbalanced?
 - What percentage of passengers belong to each class?
2. Perform Random train-test splitting and stratified train-test splitting. Compare the class distributions in the training and testing datasets.
 - a. Which splitting method better preserves the original class distribution?
 - b. Why is stratified splitting preferred for classification problems?
3. Apply Random Oversampling to the training dataset. Display Class distribution before oversampling and Class distribution after oversampling.
 - a. What changes do you observe after oversampling?
 - b. What is one advantage and one disadvantage of Random Oversampling?
4. Apply Random Undersampling to the training dataset. Display Class distribution before and after undersampling.
 - a. What changes do you observe after undersampling?
 - b. What is one advantage and one disadvantage of Random Undersampling?
5. Compare the following approaches:
 - **Original Dataset**
 - **Random Oversampling**
 - **Random Undersampling**

- a. Which technique increases the size of the dataset?
 - b. Which technique decreases the size of the dataset?
 - c. Which technique may lead to overfitting?
 - d. Which technique may result in loss of useful information?
 - e. Which technique would you recommend for this dataset? Justify your answer.
6. Apply SMOTE (Synthetic Minority Over-sampling Technique) to the training dataset and display before and after Class distributions.
 - How is SMOTE different from Random Oversampling?
 - Why is SMOTE generally preferred over simply duplicating minority class samples?